A LIMIT OF TOPOLOGICAL GROUPS WITH THE DESCENDING CHAIN CONDITION IS CLOSED IN THE PRODUCT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the limit of an inverse system of groups as the set of compatible families. Used inside the counted proof (source lines 5982-5990). Identifies the limit with a subset of the product defined by compatibility of the transitions.

Source reference. Original reference categories-lemma-limits-products-equalizers: sources/stacks-topology/pinned-original-source.tex, source lines 2582–2582. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-limits-products-equalizers, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *The category of topological groups has limits and limits commute with the forgetful functors to (a) the category of topological spaces and (b) the category of groups.*

Proof. It is enough to prove the existence and commutation for products and equalizers, see Categories, Lemma categories-lemma-limits-products-equalizers. Let G_i, $i \in I$ be a collection of topological groups. Take the usual product $G = \prod G_i$ with the product topology. Since $G \times G = \prod (G_i \times G_i)$ as a topological space (because products commutes with products in any category), we see that multiplication on G is continuous. Similarly for the inverse map. Let $a, b : G \to H$ be two homomorphisms of topological groups. Then as the equalizer we can simply take the

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equalizer of a and b as maps of topological spaces, which is the same thing as the equalizer as maps of groups endowed with the induced topology. $\square$